

Smart Cat Litter Box Monitor — Team Budget

TECHIN 515 Spring 2026 · Mengqi Shi | Yuna Xiong | Xin Luo · Team Budget: \$350.00 · Updated 2026-05-11

Hardware

| Item                                       | Spec / Source                                | Qty   | Unit (\$) | Total (\$) | Status                        |
|--------------------------------------------|----------------------------------------------|-------|-----------|------------|-------------------------------|
| XIAO ESP32S3                               | Maker Space                                  | 2     | —         | 0.00       | MAKER SPACE                   |
| Load cell, 4-bar                           | Maker Space                                  | 1     | —         | 0.00       | MAKER SPACE                   |
| HX711 amplifier module                     | Maker Space                                  | 1     | —         | 0.00       | MAKER SPACE                   |
| PIR motion sensor                          | Maker Space                                  | 1     | —         | 0.00       | MAKER SPACE                   |
| LiPo battery, 3.7 V 500 mAh                | Jameco / Amazon                              | 1     | 5.83      | 5.83       | PURCHASED                     |
| LiPo charger module (TP4057, 5pcs)         | HiLetgo / Amazon                             | 1 pk  | 7.56      | 7.56       | PURCHASED                     |
| Resistors / capacitors / headers           | Maker Space                                  | —     | —         | 0.00       | MAKER SPACE                   |
| Jumper wires & breadboard                  | Maker Space                                  | 1 set | —         | 0.00       | MAKER SPACE                   |
| Cat litter box (Smart Paws, enclosed)      | Amazon                                       | 1     | 43.99     | 43.99      | PURCHASED                     |
| XIAO ESP32S3 Sense (OV2640 camera)         | Amazon Business — Order #111-2440302-5981862 | 1     | 23.51     | 23.51      | PURCHASED<br>Delivered May 1  |
| SanDisk 32GB microSDHC (A1, C10)           | Amazon Business — Order #111-2440302-5981862 | 1     | 20.80     | 20.80      | PURCHASED<br>Delivered Apr 30 |
| ELEGOO Dupont wires 120pcs (M-F, M-M, F-F) | Amazon Business — Order #111-2440302-5981862 | 1 pk  | 6.88      | 6.88       | PURCHASED<br>Delivered Apr 30 |

PCB & Fabrication

| Item                              | Spec / Source    | Qty | Unit (\$) | Total (\$) | Notes     |
|-----------------------------------|------------------|-----|-----------|------------|-----------|
| PCB fabrication, Run 1            | School sponsored | 5   | —         | 0.00       | SPONSORED |
| PCB fabrication, Run 2 (revision) | School sponsored | 5   | —         | 0.00       | SPONSORED |

Cloud & Software

| Item                       | Details                          | Cost (\$) |
|----------------------------|----------------------------------|-----------|
| MQTT broker (HiveMQ Cloud) | Free tier — up to 10 connections | 0.00      |
| Edge Impulse               | Free tier                        | 0.00      |
| Estimated Total Spend      |                                  | \$108.57  |
| Team Budget                |                                  | \$350.00  |
| Remaining (contingency)    |                                  | \$241.43  |